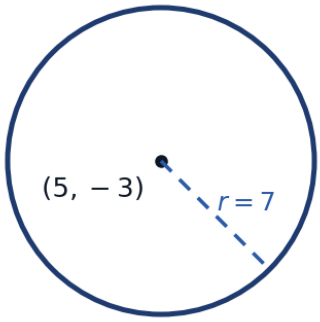
Circles, Radians, Trig & Conditional Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Standard Equation of a Circle

Find the center (h, k) and radius r of the circle $(x-5)^2 + (y+3)^2 = 49$, and state the coordinates of the four points on the circle that lie horizontally or vertically from the center.



2 [Foundational] • [FRQ] • Target: Completing the Square for Circle Equations

Convert the general circle equation $x^2 + y^2 - 8x + 6y - 11 = 0$ into standard form by completing the square, and find its center and radius.

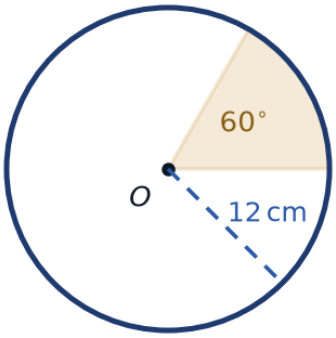
3 [Foundational] • [FRQ] • Target: Degree-Radian Conversions

Convert (a) 135° and 210° to radians, and (b) $\frac{5\pi}{6}$ and $\frac{7\pi}{4}$ radians to degrees.

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4 [Foundational] • [FRQ] • Target: Arc Length & Sector Area

In a circle of radius $r = 12\,\text{cm}$ shown below, a central angle measures $\theta = 60^\circ$ ($\frac{\pi}{3}$ rad). Find the exact arc length s and the exact area A of the shaded sector in terms of π.



5 [Foundational] • [FRQ] • Target: Inscribed Angle Theorem

On a circle with center O, an inscribed angle $\angle ABC$ intercepts minor arc AC. If $\angle ABC = 38^\circ$, find the degree measure of minor arc AC and central angle $\angle AOC$.

6 [Foundational] • [FRQ] • Target: Unit Circle Trigonometry

Evaluate the exact values of (a) $\sin\left(\frac{2\pi}{3}\right)$, (b) $\cos\left(\frac{5\pi}{4}\right)$, and (c) $\tan\left(\frac{11\pi}{6}\right)$ using reference angles on the unit circle.

7 [Foundational] • [FRQ] • Target: Tangent-Radius Perpendicularity

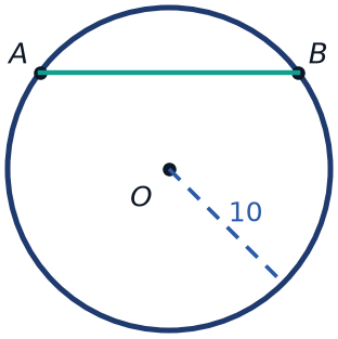
Point P is located $17\,\text{cm}$ from the center O of a circle of radius $8\,\text{cm}$. A line through P is tangent to the circle at point T. Find the length of the tangent segment PT.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

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8 [Intermediate] • [FRQ] • Target: Chord Bisector & Distance from Center

In a circle of radius $r = 10\,\text{cm}$, a chord AB has length $16\,\text{cm}$, as shown. Find the perpendicular distance d from the center O to chord AB.



9 [Intermediate] • [FRQ] • Target: Equation of a Tangent Line to a Circle

Find the equation of the tangent line to the circle $(x-1)^2 + (y+2)^2 = 25$ at the point $T(4, 2)$ on the circle.

10 [Intermediate] • [FRQ] • Target: Pythagorean Identity & Quadrant Signs

If $\cos\theta = -\frac{5}{13}$ and θ lies in Quadrant III ($\pi < \theta < \frac{3\pi}{2}$), find the exact values of $\sin\theta$ and $\tan\theta$.

11 [Intermediate] • [FRQ] • Target: Relating Sector Area & Arc Length ($A = \frac{1}{2}rs$)

A circular sector has arc length $6\pi\,\text{cm}$ and area $27\pi\,\text{cm}^2$. Find the radius r of the circle and the central angle θ in both radians and degrees.

12 [Intermediate] • [FRQ] • Target: Non-Monic Circle Equations

Find the diameter and area of the circle whose equation is $2x^2 + 2y^2 + 12x - 8y - 24 = 0$.

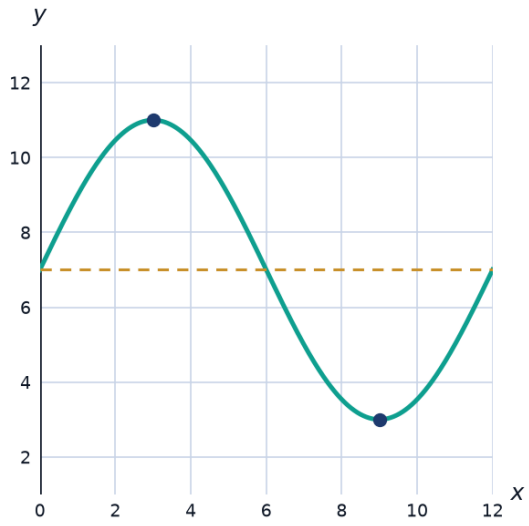
13 [Intermediate] • [FRQ] • Target: Cyclic Quadrilaterals

Quadrilateral $ABCD$ is inscribed in a circle (cyclic quadrilateral). If $\angle A = (2x + 15)^\circ$ and $\angle C = (3x - 10)^\circ$, find x, $\angle A$, and $\angle C$.

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14 [Intermediate] • [FRQ] • Target: Sinusoidal Function Parameters

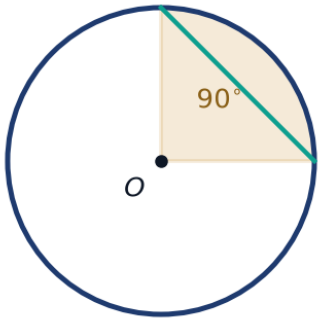
State the amplitude, period, midline, and maximum value of the sinusoidal function $f(t) = 4\sin\left(\frac{\pi}{6}t\right) + 7$, as graphed below.



Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Area of a Circular Segment • Cross-Lesson: Topic 8.01 (Composite Areas)

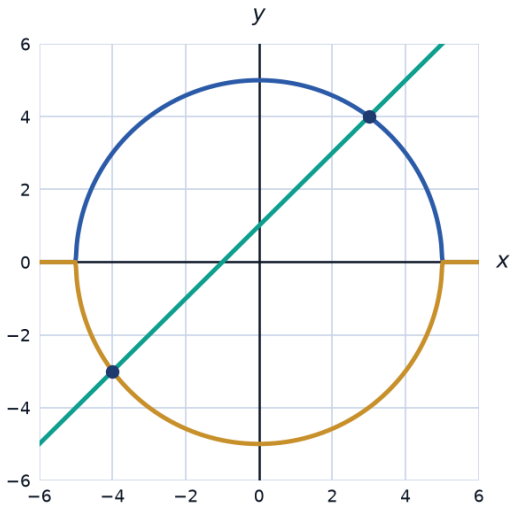
Find the exact area of the circular segment (the region between chord AB and minor arc AB) in a circle of radius $r = 6$ cm if the central angle subtended by AB is 90°.



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16 [Challenging] • [FRQ] • Target: Line-Circle Secant Intersection System • Cross-Lesson: Topic 3.01 (Quadratics) & Topic 2.01 (Systems)

Find the coordinates of all intersection points of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$, and compute the length of the secant chord.



17 [Challenging] • [FRQ] • Target: Power of a Point (Tangent-Secant Theorem) • Cross-Lesson: Topic 7.01 (Similar Triangles)

From an external point P, a tangent segment PT of length 12 and a secant line PAB (passing through the circle at A and B) are drawn. If the external secant part is $PA = 8$, find the length of the internal chord AB.

18 [Challenging] • [FRQ] • Target: Periodic Trigonometric Modeling • Cross-Lesson: Topic 4.01 (Function Modeling)

A Ferris wheel has a diameter of $40\,\text{m}$, its lowest point is $4\,\text{m}$ above the ground, and it completes one full revolution every 60 seconds. If a rider boards at the lowest point at $t = 0$, write a cosine function $h(t)$ for the rider's height above the ground at time t seconds, and find $h(15)$ and $h(30)$.

19 [Challenging] • [FRQ] • Target: Tangent Line Parameter via Point-to-Line Distance • Cross-Lesson: Topic 1.01 (Linear Equations & Absolute Value)

For what values of c is the line $3x + 4y = c$ tangent to the circle $x^2 + y^2 - 2x + 4y = 20$?

20 [Challenging] • [FRQ] • Target: Common External Tangent of Two Circles • Cross-Lesson: Topic 7.01 (Pythagorean Theorem)

Two circles of radii $r_1 = 9\,\text{cm}$ and $r_2 = 4\,\text{cm}$ are externally tangent to each other and both are tangent to the same straight line ℓ. Find the distance between the two points of tangency on line ℓ.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

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21 [Foundational] • [MCQ] • Target: Reading Circle Radius from Standard Form

In the xy-plane, what is the radius of the circle $(x + 6)^2 + (y - 8)^2 = 100$?

a. 6

b. 8

c. 10

d. 100

22 [Foundational] • [MCQ] • Target: Radians to Degrees

An angle has a measure of $\frac{5\pi}{12}$ radians. What is the measure of the angle in degrees?

a. 60°

b. 75°

c. 80°

d. 105°

23 [Foundational] • [MCQ] • Target: Arc Length Formula $s = r\theta$

In a circle with radius 8, what is the length of an arc intercepted by a central angle of $\frac{3\pi}{4}$ radians?

a. 3π

b. 6π

c. 12π

d. 24π

24 [Foundational] • [MCQ] • Target: Testing Points on a Circle

Which of the following points lies on the circle $(x - 2)^2 + (y + 1)^2 = 25$?

a. $(2, 5)$

b. $(5, 3)$

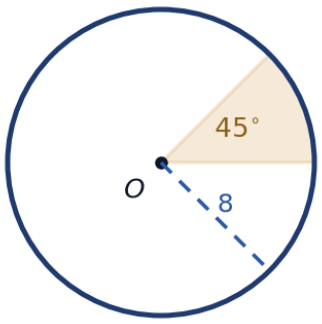
c. $(-2, 3)$

d. $(6, 2)$

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25 [Foundational] • [MCQ] • Target: Sector Area as Fraction of Circle

A circle has circumference 16π. What is the area of a 45° sector of this circle?



a. 2π

b. 4π

c. 8π

d. 16π

26 [Foundational] • [MCQ] • Target: Quadrant II Cosine Sign

If $\sin\theta = \frac{4}{5}$ and θ is in Quadrant II, what is the value of $\cos\theta$?

a. $-\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$

d. $-\frac{3}{4}$

27 [Foundational] • [MCQ] • Target: Finding Center by Completing the Square

What is the center of the circle given by $x^2 + y^2 - 10x + 4y = 7$?

a. $(-5, 2)$

b. $(5, -2)$

c. $(-10, 4)$

d. $(10, -4)$

28 [Foundational] • [MCQ] • Target: Inscribed Angle vs. Central Angle

Points $A, B,$ and C lie on a circle with center O. If central angle $\angle AOC = 110^\circ$ and B lies on the major arc AC, what is the measure of inscribed angle $\angle ABC$?

a. 55°

b. 70°

c. 110°

d. 125°

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Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 **[Intermediate]** • **[MCQ]** • **Target: Finding Constant c from Circle Radius**

The equation $x^2 + y^2 + 6x - 14y + c = 0$ represents a circle with radius 4. What is the value of the constant c?

a. 34

b. 42

c. 50

d. 74

30 **[Intermediate]** • **[MCQ]** • **Target: Circle Equation from Diameter Endpoints**

In the xy-plane, segments AB is a diameter of a circle. If $A = (-3, 2)$ and $B = (5, 8)$, what is the equation of the circle?

a. $(x - 1)^2 + (y - 5)^2 = 25$

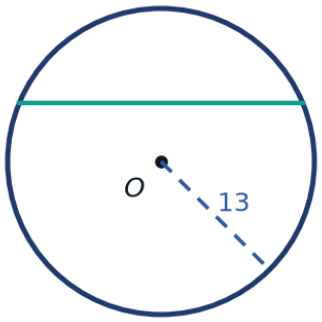
b. $(x - 1)^2 + (y - 5)^2 = 100$

c. $(x + 1)^2 + (y + 5)^2 = 25$

d. $(x - 4)^2 + (y - 3)^2 = 25$

31 **[Intermediate]** • **[MCQ]** • **Target: Chord Length from Distance to Center**

In a circle of radius 13, a chord lies at a perpendicular distance of 5 from the center. What is the length of the chord?



a. 12

b. 18

c. 24

d. 26

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32 [Intermediate] • [MCQ] • Target: Trigonometric Ratios Across Quadrants

For a curriculum assessment problem, suppose if $\tan\theta = -\frac{8}{15}$ and $\sin\theta > 0$, what is the value of $\cos\theta$?

a. $-\frac{15}{17}$

b. $-\frac{8}{17}$

c. $\frac{15}{17}$

d. $\frac{8}{17}$

33 [Intermediate] • [MCQ] • Target: Chord of Outer Circle Tangent to Inner Circle

Two concentric circles have radii 6 and 10. A chord of the larger circle is tangent to the smaller circle. What is the length of this chord?

a. 8

b. 12

c. 16

d. $2\sqrt{34}$

34 [Intermediate] • [MCQ] • Target: Angular Separation in Radians

What is the radian measure of the smaller angle between the hour hand and the minute hand of an analog clock at exactly 4:00?

a. $\frac{\pi}{3}$

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{5\pi}{6}$

35 [Intermediate] • [MCQ] • Target: Rolling Wheel Linear Distance

A wheel of radius 0.4 m rolls forward without slipping through 15 complete revolutions. How far does the center of the wheel travel?

a. 6π m

b. 12π m

c. 15π m

d. 24π m

36 [Intermediate] • [MCQ] • Target: Circle Tangency to Coordinate Axes

If the circle $(x-4)^2 + (y-k)^2 = 25$ is tangent to the x-axis, what are the possible values of k?

a. ± 4

b. ± 5

c. ± 9

d. ± 25

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37 [Intermediate] • [MCQ] • Target: Simplifying Trigonometric Expressions

Which expression is equivalent to $\frac{1-\cos^2\theta}{\sin\theta\cos\theta}$ for all θ where the expression is defined?

a. $\sin\theta$

b. $\cos\theta$

c. $\tan\theta$

d. $\frac{1}{\tan\theta}$

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Extremal Distance from a Point to a Circle • Cross-Lesson: Topic 7.01 (Distance Formula)

What is the greatest distance from the point $P(7, 10)$ to any point on the circle $(x-1)^2+(y-2)^2=9$?

a. 10

b. 13

c. 16

d. 19

39 [Challenging] • [MCQ] • Target: Intersecting Chords Theorem & Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Equations)

Two chords AB and CD intersect at point E inside a circle. If $AE = 4$, $EB = 9$, and $CE = x$, $ED = x + 5$, what is the length of chord CD?

a. 4

b. 9

c. 13

d. 15

40 [Challenging] • [MCQ] • Target: Coordinate Intercepts & Thales' Diameter Theorem • Cross-Lesson: Topic 1.01 (Intercepts) & Topic 7.01 (Right Triangles)

In the xy-plane, the circle $x^2+y^2-6x-8y=0$ intersects the x-axis at the origin $(0, 0)$ and at point A, and intersects the y-axis at the origin and at point B. What is the length of segment AB?

a. 5

b. 7

c. 10

d. 14

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41 [Challenging] • [MCQ] • Target: Pulley Belt Composite Perimeter • Cross-Lesson: Topic 8.01 (Composite Perimeter)

A belt wraps tightly around two congruent circular pulleys, each of radius $4\,\text{cm}$, whose centers are $14\,\text{cm}$ apart. What is the total length of the belt?

a. $14 + 8\pi$

b. $28 + 4\pi$

c. $28 + 8\pi$

d. $28 + 16\pi$

42 [Challenging] • [MCQ] • Target: Trigonometric Equations Quadratic in Form • Cross-Lesson: Topic 3.01 (Quadratic Factoring)

For how many values of x in the interval $0 \le x \le 2\pi$ does $2\sin^2(x) - \sin(x) - 1 = 0$ hold?

a. 1

b. 2

c. 3

d. 4

43 [Challenging] • [MCQ] • Target: Incircle of an Equilateral Triangle • Cross-Lesson: Topic 7.01 (Equilateral Triangles)

A circle is inscribed in an equilateral triangle of side length 12. What is the area of the circle?

a. 6π

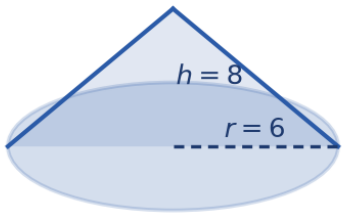
b. 12π

c. 18π

d. 36π

44 [Challenging] • [MCQ] • Target: Sector-to-Cone 3D Reconstruction • Cross-Lesson: Topic 8.01 (3D Cone Volume)

When a circular sector with central angle 216° and radius 10 is folded into a right circular cone, what is the volume of the cone?



a. 96π

b. 128π

c. 144π

d. 288π

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45 [Challenging] • [MCQ] • Target: Intersecting Circles & Rhombus Area • Cross-Lesson: Topic 8.01 (Rhombus Area)

If the circles $x^2 + y^2 = 9$ and $(x - 4)^2 + y^2 = 9$ intersect at points P and Q, what is the area of the rhombus $OPO'Q$ formed by the two centers $O(0, 0), O'(4, 0)$ and the intersection points P, Q?

a. $4\sqrt{5}$

b. $6\sqrt{5}$

c. $8\sqrt{5}$

d. 12

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